

$$\begin{bmatrix} 9 & 15 \\ 15 & 50 \end{bmatrix} = F \bullet F^T$$

$$= L L^T$$

$$L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}$$

$$\text{for } l_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} l_{ik}^2}$$

$$\Rightarrow l_{11} = \sqrt{a_{11}} = 3$$

~~l_{22}~~

$$l_{22} = \sqrt{a_{22} - \sum_{k=1}^1 l_{2k}^2}$$

$$l_{22} = \sqrt{a_{22} - l_{21}^2}$$

$$l_{22} = \sqrt{25}$$

$$\boxed{l_{22} = 5}$$

$$\text{for } l_{ij} = \frac{1}{l_{jj}} \left(a_{ij} - \sum_{k=1}^{j-1} l_{ik} l_{jk} \right)$$

$$l_{21} = \frac{1}{l_{11}} \left((a_{21}) + \cancel{0} \right)$$

$$= \frac{15}{3}$$

$$\boxed{l_{21} = 5}$$

$$\therefore L = \begin{bmatrix} 3 & 0 \\ 5 & 5 \end{bmatrix}$$